

W1 | How School Resources Are Used

La Cisterna · Group 03 · September 25, 2026

Members: Javier Alcaíno, Lucas Riquelme, and Renato Varela.

The group reported instructor authorization to use the educational topic instead of the health data suggested by the brief. W1 has not yet been submitted and has not received feedback on this analysis.

Decision, problem, and rationale

Main question: what public resources do public and subsidized-private schools in La Cisterna receive and account for, what expenses show documented findings, and what evidence proves delivery and use? The comparison of educational results is kept as context and never as standalone proof of misuse.

Relevance goes beyond the datasets themselves: DIPRES's evaluation of the Preferential School Subsidy (**DIPRES, March 2023**) examined resource use and accountability processes; **Law 20.248**, current text consulted on 25-09-2026, links the SEP subsidy to PME actions and requires reporting its use to the Superintendency and the community [1, 2]. Mineduc's **2025 PME technical guidelines** define that plan as the official tool with which a school plans and accounts for its improvement, which grounds line B beyond the statistical databases [3]. The territorial decision is to reconstruct that chain for each RBD, without assuming that an anomaly equals a violation.

Line	Proposed user and use	Initial question
A · Accountability and oversight	Superintendency and school boards: select case files, correct financial reports, and verify findings.	Which declared expense was accepted, rejected, or subject to repayment, and why?
B · School Improvement Plan (PME) and Quality Agency	School leadership teams and school communities: compare declared planning, coverage, and relevant results.	What registration or implementation gaps require additional evidence?

Provisional preference: A directly addresses resource use. B helps define questions and comparisons. Detailed financial reports and case files are not yet available, so W1 demonstrates feasibility and limits, not quantified misspent money.

First data inspection

The Directory reports 60 RBDs in 2022 and 2023, and 59 in 2024 and 2025. The eight public schools appear as DAEM through 2024 and as SLEP in 2025; RBD 9830 appears only in the first two years. RBD 9860, under delegated administration, was excluded from the census because it belongs to a third dependency type outside the requested scope (public and subsidized private); its exclusion holds across all four cycles and is recorded in `data/PROCEDENCIA.md`. The 2024 PME extract records 767 actions from 46 RBDs. The figure counts schools with at least one row, not eligible coverage or executed spending.

PME 2024: schools with at least one action in the extract

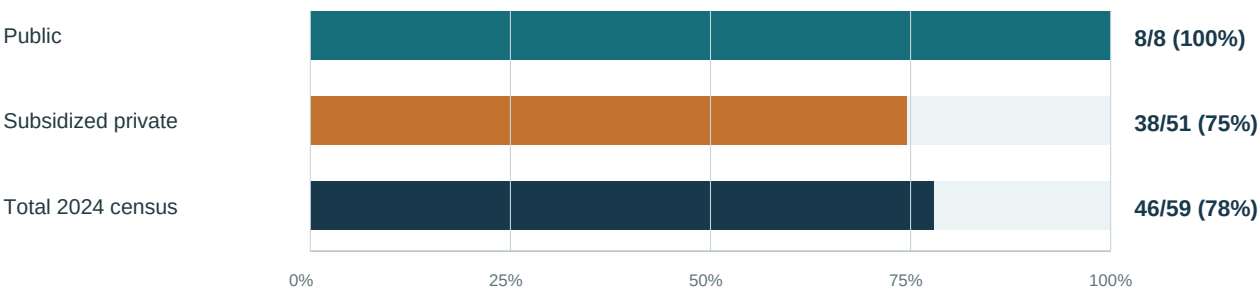


Figure 1. The PME extract includes 8/8 public schools and 38/51 subsidized-private schools from the 2024 census (46/59 overall, 0-100% axis for direct reading). The 13 without a row require checking applicability and source; they are not classified as non-compliance. Source: 2024 Official Directory and PME 2024, calculated in `analysis/outputs/metricas.json`.

1. Inspected data and dictionary

Source and row	Proposed key before joining	Observed result and limit
Directory: RBD-year (2022-2025)	`anio + rbd`	238 rows, 60/60/59/59. Offer, dependency, and enrollment define the annual census.
PAS: process (2022-2025 files)	`pa_id`	110 processes across 39 RBDs; 47 fines and 0 repayments at first instance. Financial subject matter not available in the extract.
PME: declared action 2024	`fila_excel` is an extraction locator, not an official ID	767 actions, 46 RBDs; total estimated amount \$9,009,408,718, not a payment.
SIMCE: RBD-year-grade	`anio + rbd + grado`	236 rows. A score can be missing even with a row present.
IDPS: RBD-year-grade-indicator	`anio + rbd + grado + indicador`	936 rows. A missing average is not zero.

The `data/DICCIONARIO.md` dictionary defines fields, periods, and units; `analysis/outputs/perfil\_fuentes.csv` reports actual nulls and duplicates. The keys in this table were **proposed**; the pipeline first profiled them and then checked uniqueness, membership, and conservation in `Validate`. No join was assumed safe just for sharing the RBD name.

2. Analytical preparation: eight stages

Stage	Action and evidence it leaves behind
Load	Reads five extracts with no additional filters; 238 + 110 + 767 + 236 + 936 rows.
Profile	Records the proposed key, critical nulls, and duplicates per source (`perfil_fuentes.csv`).
Explore	Counts the census by year and PAS by file before joining (`controles_etapas.csv`).
Clean	Converts RBD/year to integers, amounts to Decimal, and grades to a uniform code; no rows are dropped.
Impute	Keeps `NA` for spending, reporting, and scores. A zero for PAS/PME is a count of published rows, flagged as present.
Transform	Aggregates each source before the join; builds 238 RBD-year rows (`panel_rbd_anual.csv`).
Validate	Checks unique keys, annual membership, conserved totals, and zero loss from the join.
Explore again	Re-reads 60/60/59/59 and 8/8 against 38/51 after validation; distinguishes absence of a record from absence of spending.

Second look: differences within the PME

Dimension	Actions	Declared 100%	Zero-estimate
School Climate	195	114	32
Pedagogical Management	237	147	50
Resource Management	176	112	24
Leadership	159	96	43

Of 767 actions, 469 were declared complete and 149 have a total estimate of zero. The highest median estimate per action is Resource Management (\$11,775,000). These are *declared* statuses and amounts; the table does not measure financial execution or service quality.

Data quality controls and limits

Five data-quality controls (Five C's, confirmed against the Class 03 material): **Clean** - data must be free of invalid or impossible values, which would otherwise distort statistical summaries and analytical models (negative amounts, RBDs outside the census); **Consistent** - standardizing labels and keys by RBD and year prevents inconsistent naming or joins from creating artificial subgroups; **Complete** - essential information must not be missing, as incomplete records can alter the target population and lead to biased conclusions; **Current** - information must be up to date and aligned with the specific 2022-2025 time window the decision requires (version and extraction date); **Collaborative** - datasets and definitions must be understandable across the three team members and consistent with `data/DICCIONARIO.md`, so the analysis is not analytically fragile or siloed in whoever built it. Documentary traceability is assessed separately, in the Process record.

3. Critical comparison to decide C1

Brief dimension	A · Accountability / PAS	B · PME / Agency
Relevance supported	SEP Law and the 2023 DIPRES evaluation [1,2]: resource use and accountability are part of the public design.	Mineduc's 2025 PME technical guidelines and the SEP Law [2,3]: the PME is the official, evidence-based planning tool that accounts for educational improvement.
Fit between question and inspected data	PAS shows process, instance, and sanction; accepted/rejected spending by account is missing.	PME measures declarations; SIMCE/IDPS by grade provide context, they do not verify payment or use.
Analytical opportunity for C1	Cross-reference accountability records, case files, and delivery documents; compare cases with and without findings.	Quality and coverage panel by RBD/year/grade; cases to request supporting evidence for actions.
Main limitation/uncertainty	PAS is not a census of audits nor of misspent amounts. The 2025 closing of accounts depends on availability.	PME applicability and grade-level eligibility are not yet resolved; there is no educational causality.
Feasible scope; ML and dashboard	A documentary dashboard now. ML only with labels for final resolutions, exposure, and reviewed negatives.	A coverage dashboard now. ML only with several cohorts, comparable results, and validated statuses.

Future potential, conditional on new data

A - ML: classify whether a financial report will have a *final* non-accepted expense, using data prior to closing, account/subsidy, history, and dated administrative labels. Financial reports, the audited universe, and reviewed negative cases are still missing; no model is trained today. **Dashboard A:** for the Superintendency/school boards, RBD, year, subsidy, declared/accepted/rejected amounts, appeal status, case file, and missing document.

B - ML: estimate the risk that a PME action will not complete its declared implementation in the next cycle, using a multi-year panel of actions with verified statuses. Comparable history, a stable definition of an action, and validation of self-reported data are still missing. **Dashboard B:** for leadership teams/community, PME presence, actions by dimension, SIMCE/IDPS results by grade and version, with flags for non-applicability and missing data.

4. Seven steps of question formulation (Class 03)

Step	Application to La Cisterna
1 Decision	Choose where to review documents and which claim can be published.
2 Success	Visible coverage and findings with amount, status, and support, with no false zeros.
3 Business question	Which received, accounted-for, and flagged resources reached a verifiable service?
4 Process	Transfer -> accountability report -> oversight -> resolution -> receipt/use.
5 Entities/events/states	RBD, school board, subsidy, account, action, PAS case, document, appeal status.
6 Representation	RBD-year as a panel; accounts and case files in separate tables; a bridge for shared purchases.
7 Measurement	Final accepted/rejected amount, coverage by source, balance, and proven delivery/use.

The seven steps formulate the problem; the eight stages on the previous page prepare the data. The question is not resolved by a correlation between PME spending and SIMCE scores.

5. Process record

Decisions and checks

Date/stage; decision	Alternative, reason, and evidence	Check / result / file
25-09-2026 - scope: La Cisterna and both dependency types	Discarded following an external case; this answers the group's territorial question. 2022-2025 directories [4].	60/60/59/59 RBDs, RBD 9830 only in 2022-23. <code>`data/universo_la_cisterna_2022_2025.json`</code> ; <code>`outputs/universo_anual.csv`</code> .
25-09-2026 - two lines A/B; A preferred	Alternative: PME as proof of misuse. Rejected because these are declared estimates [5].	767 actions from 46 RBDs; not payments. <code>`outputs/metricas.json`</code> ; pages 1 and 3.
25-09-2026 - join by RBD-year after profiling	Alternative: a direct RBD join, which would multiply actions and Agency rows.	Unique keys, 238 rows conserved; <code>`outputs/perfil_fuentes.csv`</code> , <code>`outputs/controles_etapas.csv`</code> , <code>`outputs/panel_rbd_anual.csv`</code> .
25-09-2026 - no imputing of spending/scores	Alternative: fill missing values with zero, which would produce false evidence of inactivity or low results.	Presence flags and <code>`NA`</code> conserved; <code>`analysis/pipeline_la_cisterna.py`</code> , Impute stage.

Instructor feedback and response

Actual status as of 25-09-2026: W1 has not yet been submitted; no instructor feedback exists on the two lines, the figures, or the methodology. The prior authorization of the educational topic is recorded as a scope decision, not as an evaluation. The questions requesting feedback are at the end of this report. There is no advice accepted, adapted, or rejected to attribute to the professor.

Individual contributions

Member	Contribution reported by the group
Javier Alcaíno	Part of the implementation and debugging of the analytical code; structure, drafting, and review of the report.
Lucas Riquelme	Critical review of sources and institutional context; precision of data scope and conclusions.
Renato Varela	Preliminary research on the core of the project; data analysis and code review.

These contributions were confirmed by the group for this review. Verifiable work locations in the package are ``analysis/``, ``data/``, and ``report/``; the package does not contain an individual edit history to attribute specific lines of code.

AI use and verification

Purpose and prompt excerpt	Output and human/analytical decision	Verification
Asked for help narrowing the territorial scope back to La Cisterna and building the annual census across both dependency types. Codex helped organize the census tables and the join structure.	The annual census was accepted; the fixed cutoff of 59 was revised after detecting RBD 9830 in 2022-23. Extrapolating 2025 back to earlier years was not accepted.	Original directory, <code>`data/universo...2022_2025.json`</code> , <code>`outputs/universo_anual.csv`</code> .
Asked for help structuring the data-preparation pipeline into clear, checkable stages. Codex drafted an initial version of the stage sequence.	The eight stages were accepted. Imputing amounts and scores with no basis was rejected. Proposed keys were validated before the panel.	<code>`analysis/pipeline_la_cisterna.py`</code> , <code>`outputs/controles_etapas.csv`</code> , <code>`outputs/perfil_fuentes.csv`</code> .
Asked for help checking consistency between the computed figures, the notebook, and the report draft. Codex helped regenerate the analysis outputs, notebook, and document layout after each correction.	110 PAS, 767 PME, and 236/936 Agency rows were accepted after checks. The row/score distinction was corrected.	<code>`analysis/W1_analisis_ejecutado.ipynb`</code> , <code>`outputs/metricas.json`</code> ; visual render of the PDF.

Individual reflection

Lucas Riquelme

My initial understanding of the system was basic: there are public and subsidized-private schools, and they receive additional funding earmarked for vulnerable students, whether due to social vulnerability or poverty. The idea behind the project was to use Business Intelligence to cross-reference that information and see whether the funding was being used properly.

As the work progressed, however, I had to correct a bias I started with. My initial hypothesis was that, since these were state funds, there was probably some element of corruption or misuse behind the figures. Digging deeper, I understood two things that dismantle that hypothesis as I had originally framed it. First, that with the data we have there is no mathematical way to establish that causality; the PME, for example, consists of estimates declared by the school itself, not proof of payment, and the PAS documents sanction processes, not a registry of misappropriated spending. Second, and this was the most relevant point for me, that Business Intelligence does not aim to prove a hypothesis the way inferential statistical analysis would. It is not about concluding "this is corruption," but about showing patterns and evidence of sufficient quality for someone — the Superintendency, a leadership team — to make an informed decision.

That shift in approach redirected my conclusion toward something more applicable: rather than pursuing a conclusive claim about resource misuse, what is called for is defining a monitoring KPI — for example, the year-over-year variation in PME actions with no execution or no declared estimate — that would allow management or leadership teams to be alerted about which schools need attention, without accusing anything the data cannot support.

Finally, when technically evaluating the two lines of work we proposed, I identified a trade-off we had not made explicit. Line B (PME and Agency) answers our business question more directly and specifically, because its data — declared actions by dimension, estimates, coverage — fits better with what we are asking. However, that data is self-reported by the school itself. Line A (Accountability and PAS), on the other hand, has less coverage relative to the question — it does not reach accepted or rejected spending by account — but it originates in the Superintendency of Education's oversight and sanction processes, that is, in a source external to the school itself, which gives it a different degree of reliability compared to self-reported information. This leads me to conclude that neither line is absolutely superior: specificity to the question must be weighed against reliability of the source, and the project's final answer will likely need to combine both rather than discard one.

Renato Varela

My case was particular, since we began investigating this topic after having learned about some real instances of misspent money. That is what makes this course interesting: it pushes you to reflect on how to actually prove something, or investigate it methodically.

What strikes me is the whole chain through which resources are assigned, verified, and managed — moving from the SEP subsidy to the PME, and on to reporting and oversight in case of inconsistencies. Throughout this process there are criteria so subjective that they complicate the analysis and make it hard to say, unequivocally, "this is corruption." Something that looks so obvious when you look at reality turns out to be so challenging in theory, and that is exactly what makes it worth investigating. Part of why we chose La Cisterna was precisely this prior knowledge of real cases. We could have picked other municipalities with expenses that were categorically poorly reported, but being able to talk to people who know the education field, plus the visible example of "questionable investments," made this municipality especially interesting.

After investigating, we noticed a wide margin for improvement, which is why interviewing is our best next option. For example, we see 47 fines within the PAS (administrative sanction processes). On its own that number stands out, but it is not conclusive given the zero repayments recorded at first instance. It takes only a little creativity and context to realize that these cases could vary slightly and get registered as a new instance rather than as repeated conduct. This hypothetical case would need to be investigated; we still cannot claim it. Moreover, even assuming these fines occur in categorically different contexts (no trick involved), that does not mean there is accepted, rejected, or misused spending. All of this data is useful for prioritizing which schools' financial reports we should investigate in more detail.

Javier Alcaíno

My initial approach to the project was fairly technical. At first I thought the main challenge would be obtaining the databases, joining them, and finding some interesting pattern among school resources, PME actions, and their results. As we made progress, I realized that having a lot of data does not necessarily mean having comparable information, and that an important part of the work happens even before the analysis begins.

This became clear when it came time to code and join the different databases. Not all of them shared the same unit of analysis: some were by school and year, others contained individual PME actions, and others administrative sanction processes. If we had simply joined them using the RBD, we could have ended up multiplying records and producing results that looked real but were actually the product of a bad join. That is why we ended up profiling the databases, checking the keys, and aggregating the information before building the panel. I was also struck by something as simple as the difference between a zero and a missing value. Putting a zero where there is no information can completely change the interpretation and make it look like a school did not spend, did not execute, or had a poor result, when in reality we simply do not have that data.

From that, I understood more clearly that Business Intelligence is not just about building an attractive dashboard or finding correlations, but also about building information that is traceable and that lets you know where each result came from. In our case, even after correctly processing the databases, we still cannot claim that resource misuse exists. We can show that there are PME actions, estimated amounts, sanction processes, or the absence of certain records, but each of those things has a different meaning and cannot be mixed together as if they were evidence of the same thing.

I think one of the most important results of this first stage is precisely having identified what we still lack. The current analysis lets us prioritize schools and flag cases worth reviewing, but to truly answer our question we need to go down to a more specific level: financial reports, accepted or rejected expenses, case files, and documents proving that a resource actually arrived and was used. For me, the challenge for the next stages will be to keep this same logic: that every indicator we show has a clear definition and that we can defend how it was built, instead of trying to draw a stronger conclusion than the data allows.

6. Gaps, next documents, and questions for the professor

Data not yet obtained: financial reports by RBD-year-subsidy-account for 2022-2024 and 2025 if closed; accepted/rejected spending, balance, and repayment; PAS case files with financial subject matter; invoices, receipt, and use of selected transactions. The package contains the Agency's public statistical databases, but not a collection of individual school reports. The 17 special-education schools require results relevant to their offer, without imputing a zero SIMCE score to them.

Case selection: after classifying PAS subject matter and receiving financial reports, prioritize non-accepted expenses or unaccredited balances with verifiable amount and status; include a sample with no signal to evaluate false alarms. A fine is a sanction, not a misspent amount. An unexecuted balance can be carried forward. A later school result does not prove causal use of a purchase.

Questions for feedback: 1) Is it appropriate to keep special education in the financial census and report it separately for educational results? 2) What minimum evidence does C1 require to move from a PAS record to a spending finding? 3) Is the four-year window sufficient if 2025 does not yet have a closed financial report? 4) Should B remain as context, or be evaluated as an independent alternative question?

References and official policy

[1] DIPRES (2023), *Evaluación Subvención Escolar Preferencial. Informe final* (Preferential School Subsidy Evaluation, final report), March 2023; resource use and accountability processes. [Official report](#)

[2] National Congress Library, *Ley 20.248 de Subvención Escolar Preferencial* (Preferential School Subsidy Law), text in force, consulted 25-09-2026; arts. 6-8, PME and accountability. [Legal text](#)

[3] Mineduc, General Education Division (2025), *Plan de Mejoramiento Educativo 2025: principios, orientaciones actualizadas y uso de la plataforma* (School Improvement Plan 2025: principles, updated guidelines, and platform use); defines the PME as the official, evidence-based planning and decision-making tool. [Technical document](#)

[4] Mineduc Studies Center, *Directorio Oficial de Establecimientos Educacionales* (Official Directory of Schools), 2022, 2023, 2024, and 2025 files, annual enrollment cut. [Catalog](#)

[5] Mineduc, *Bases de datos PME, Implementación PME 2024* (PME databases, 2024 Implementation), public file and description of declared estimates. [Catalog](#)

[6] Superintendency of Education, *Procesos administrativos sancionatorios* (Administrative sanction processes), 2022-2025 files and dictionary. [Official data](#)

[7] Quality Agency, *Información Estadística* (Statistical Information), public SIMCE and IDPS databases 2023-2025. [Official databases](#)

Extraction files, versions, and limits: `data/PROCEDENCIA.md`, `data/DICCIONARIO.md`, `data/fuentes.json`, and `data/manifiesto\_extraccion.json`. Figures and checks: `analysis/outputs/`. The code allows reproducing them without a network connection.